

Statistical Analysis of Arrival and Service Time Distributions for an ATM Queueing System

A Case Study of the HBL ATM, Lucky One Mall, Food Court (2nd Floor)

Course:	Modeling & Simulation
Assignment:	Distribution Fitting & Chi-Square Goodness-of-Fit Analysis
Dataset:	atm_dataset_preview.csv (n = 130 observed transactions)
Significance Level:	$\alpha = 0.05$
Tools Used:	Python (pandas, SciPy, Matplotlib)

Prepared for submission — student name, registration number, and section may be added above by the submitting student.

1. Dataset Description

The dataset used in this analysis records 130 customer transactions observed at an HBL ATM located in the Food Court on the 2nd Floor of Lucky One Mall. Each record contains a customer identifier, the cumulative arrival time (in minutes from the start of observation), the clock time of arrival, the interarrival time (time gap since the previous customer, in minutes), the service time (transaction duration at the ATM, in minutes), and the transaction type (Withdrawal, Deposit, Balance Inquiry, Fund Transfer, or Mini Statement).

This report analyzes the two variables that are central to queueing model calibration: **InterarrivalTime** and **ServiceTime**. The dataset was used exactly as provided; no values were added, removed, or modified.

Transaction Type	Count	Percentage
Withdrawal	73	56.2%
Balance Inquiry	21	16.2%
Deposit	17	13.1%
Fund Transfer	11	8.5%
Mini Statement	8	6.2%

Table 1.1 — Composition of the dataset by transaction type.

2. Methodology

For each variable, descriptive statistics were computed first (mean, median, mode, standard deviation, variance, minimum, and maximum). A theoretical distribution was then proposed based on the standard assumptions of queueing theory: interarrival times are conventionally modeled as Exponential (corresponding to a Poisson arrival process), while service times were evaluated against three common candidate distributions — Normal, Gamma, and Lognormal — to determine the best fit.

Parameters of each candidate distribution were estimated from the sample data using the Method of Maximum Likelihood Estimation (MLE). A Chi-Square Goodness-of-Fit test was then performed for the selected distribution in each case, using the following procedure:

(i) the observed data range was divided into class intervals of equal width; (ii) the observed frequency (O_i) of data points in each class was counted; (iii) the expected frequency (E_i) for each class was computed as $E_i = n \times [F(\text{upper bound}) - F(\text{lower bound})]$, where F is the CDF of the fitted distribution; (iv) adjacent classes were merged, starting from the tail of the distribution, whenever the expected frequency of a class fell below 5 — this is required because the Chi-Square approximation becomes unreliable for small expected counts; (v) the Chi-Square statistic was computed as $\sum (O_i - E_i)^2 / E_i$ over the merged classes; and (vi) this statistic was compared against the critical value from the Chi-Square distribution table at $\alpha = 0.05$ with degrees of freedom $df = k - 1 - m$, where k is the number of classes remaining after merging and m is the number of distribution parameters estimated from the data.

For the service time variable, the best-fitting candidate distribution among Normal, Gamma, and Lognormal was selected using three complementary criteria: (a) the sample skewness, to check whether a symmetric distribution such as Normal is even appropriate; (b) the Akaike Information Criterion (AIC), which rewards goodness-of-fit while penalizing model complexity; and (c) the Kolmogorov-Smirnov (K-S) statistic and p-value, which measure the maximum deviation between the empirical and fitted cumulative distributions.

3. Descriptive Statistics

Both variables are continuous, so a single unique “mode” is not statistically meaningful in the classical sense (no two raw values are expected to repeat exactly except by coincidence). Instead, the modal class — the class interval with the highest observed frequency — is reported for each variable, consistent with standard practice for grouped continuous data.

Statistic	Interarrival Time (min)	Service Time (min)
Sample Size (n)	130	130
Mean	3.3075	1.5961
Median	2.0050	1.3800
Modal Class	[0, 3) — freq. 84	[1.0, 1.5) — freq. 32
Standard Deviation	4.3910	1.0000
Variance	19.2805	0.9999
Minimum	0.00	0.30
Maximum	30.97	4.64
Skewness	3.5212	0.8720

Table 3.1 — Descriptive statistics for both variables (computed directly from the dataset).

Both variables show positive skewness (right-skewed distributions), which is expected for non-negative time measurements and rules out the Normal distribution as a natural choice for either variable on theoretical grounds — this is examined formally in Sections 4 and 5.

4. Interarrival Time Analysis — Test for Exponential Distribution

4.1 Hypotheses

H_0 : The interarrival times follow an Exponential distribution (equivalently, customer arrivals follow a Poisson process).

H_1 : The interarrival times do not follow an Exponential distribution.

Significance level: $\alpha = 0.05$

4.2 Parameter Estimation (Method of Maximum Likelihood)

For the Exponential distribution, the MLE of the rate parameter λ is the reciprocal of the sample mean:

$$\hat{\lambda} = 1 / \bar{x} = 1 / 3.3075 = 0.30234 \text{ customers per minute}$$

This implies a fitted mean interarrival time of $1/\hat{\lambda} = 3.3075$ minutes, which by construction equals the sample mean (3.3075 min) — as expected for MLE of the Exponential distribution.

4.3 Class Intervals and Frequencies (Before Merging)

The data (range 0 to 30.97 minutes) was divided into class intervals of width 3 minutes. The expected frequency for each class was computed as $E_i = n \times [F(b) - F(a)]$, where $F(t) = 1 - e^{-\lambda t}$ is the CDF of the fitted Exponential distribution, and a, b are the class boundaries.

Class Interval (min)	Observed (O)	Expected (E)
[0, 3)	84	77.5156
[3, 6)	26	31.2951
[6, 9)	12	12.6346
[9, 12)	4	5.1009
[12, 15)	1	2.0594
[15, 18)	0	0.8314
[18, 21)	1	0.3357
[21, 24)	0	0.1355
[24, 27)	1	0.0547
[27, 30)	0	0.0221
[30, 33)	1	0.0089

Table 4.1 — Raw class intervals before merging. Classes shaded in the discussion below have $E < 5$.

Classes with expected frequency below 5 are: [12, 15), [15, 18), [18, 21), [21, 24), [24, 27), [27, 30), [30, 33). Since the Chi-Square approximation is unreliable when expected frequencies are too small, these tail classes were merged successively (from the highest class downward) with their neighbor until each merged group's expected frequency reached at least 5.

4.4 Merged Class Intervals and Chi-Square Calculation

Class Interval (min)	Observed (O)	Expected (E)	$(O - E)^2 / E$
[0, 3)	84	77.5156	0.5424
[3, 6)	26	31.2951	0.8959
[6, 9)	12	12.6346	0.0319
≥ 9	8	8.5486	0.0352
TOTAL	130	129.9940	1.5054

Table 4.2 — Merged class intervals ($k' = 4$ classes) with Chi-Square contribution per class.

4.5 Test Statistic, Degrees of Freedom, and Decision

$$\text{Chi-Square statistic: } \chi^2_{\text{calc}} = \sum (O - E)^2 / E = 1.5054$$

$$\text{Degrees of freedom: } df = k' - 1 - m = 4 - 1 - 1 = 2 \quad (m = 1 \text{ parameter, } \lambda, \text{ estimated from the data})$$

$$\text{Critical value: } \chi^2_{0.05, 2} = 5.9915$$

$$p\text{-value: } 0.4711$$

Since $\chi^2_{\text{calc}} = 1.5054$ is **less than** the critical value 5.9915, we **fail to reject H_0** at the 5% significance level.

Decision: Fail to Reject H_0 . As a supplementary check (not the primary test), a Kolmogorov-Smirnov test against the fitted Exponential distribution gives $D = 0.0873$, $p = 0.2591$, which is consistent with the Chi-Square conclusion.

4.6 Conclusion

There is not enough statistical evidence, at the 5% significance level, to reject the assumption that customer interarrival times at this ATM follow an Exponential distribution with rate $\lambda \approx 0.302$ arrivals per minute (equivalently, a mean interarrival time of ≈ 3.31 minutes). This supports the standard queueing-theory assumption of Poisson arrivals for this ATM, and justifies using the Exponential distribution for the arrival process in the M/M/1, M/M/s, M/G/1, and M/G/s models used elsewhere in this project.

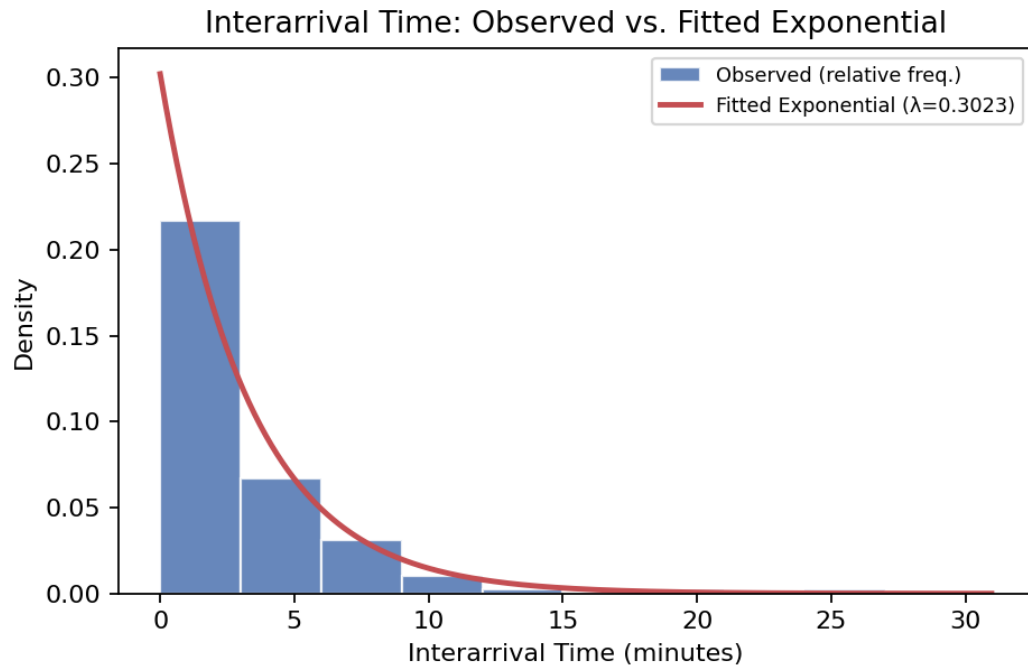


Figure 4.1 — Histogram of observed interarrival times with the fitted Exponential density overlaid.

5. Service Time Analysis — Selecting and Testing the Best-Fit Distribution

5.1 Candidate Distributions

Service time is strictly positive and its sample skewness is 0.8720 (moderately right-skewed), which already suggests a symmetric distribution is not the ideal choice. Three candidate distributions commonly used for service times in queueing analysis were fitted to the data by Maximum Likelihood Estimation: Normal, Gamma, and Lognormal.

Distribution	Fitted Parameters	AIC	K-S Statistic	K-S p-value
Normal	$\mu = 1.5961, \sigma = 0.9961$	371.91	0.1006	0.1342
Gamma	shape $k = 2.4488$, scale $\theta = 0.6518$	338.93	0.0476	0.9156
Lognormal	shape $\sigma = 0.7010$, scale $= 1.2836$	345.48	0.0770	0.4033

Table 5.1 — Comparison of candidate distributions fitted to Service Time (lower AIC and higher K-S p-value indicate a better fit).

The **Gamma distribution** gives the lowest AIC (338.93 vs. 371.91 for Normal and 345.48 for Lognormal) and the highest Kolmogorov-Smirnov p-value (0.9156), indicating it is the best-supported distribution among the three candidates for this dataset. This is also the most theoretically appropriate choice: the Gamma distribution is a natural generalization of the Exponential distribution and is the standard model for service times in M/G/1-type queues when service is "more regular" than purely random (here, shape $k = 2.45 > 1$ indicates lower variability than a pure Exponential service process, consistent with an ATM transaction having a minimum handling time before it can complete). The Normal distribution is rejected on both statistical and theoretical grounds: it assigns non-zero probability to negative service times, which is physically impossible, and it fits the right-skewed shape of the data noticeably worse (lowest K-S p-value, highest AIC).

The remainder of this section performs a formal Chi-Square Goodness-of-Fit test for the **Gamma** distribution, the selected best fit.

5.2 Hypotheses

H_0 : Service times follow a Gamma distribution with the estimated shape and scale parameters.

H_1 : Service times do not follow this Gamma distribution.

Significance level: $\alpha = 0.05$

5.3 Parameter Estimation (Method of Maximum Likelihood)

Fitting a two-parameter Gamma distribution (location fixed at 0, since service time cannot be negative) by MLE gives: shape $k = 2.4488$ and scale $\theta = 0.6518$. The implied mean is $k \times \theta = 1.5961$ minutes, matching the sample mean of 1.5961 minutes.

5.4 Class Intervals and Frequencies (Before Merging)

The data (range 0.30 to 4.64 minutes) was divided into class intervals of width 0.5 minutes. Expected frequencies were computed from the fitted Gamma CDF.

Class Interval (min)	Observed (O)	Expected (E)
[0.0, 0.5)	17	12.7162
[0.5, 1.0)	23	29.3799
[1.0, 1.5)	32	29.0894
[1.5, 2.0)	21	22.1814
[2.0, 2.5)	14	14.8985
[2.5, 3.0)	8	9.2829
[3.0, 3.5)	7	5.5036
[3.5, 4.0)	5	3.1498
[4.0, 4.5)	2	1.7557
[4.5, 5.0)	1	0.9588

Table 5.2 — Raw class intervals before merging.

Classes with expected frequency below 5 are: [3.5, 4.0), [4.0, 4.5), [4.5, 5.0). These tail classes were merged successively from the highest class downward until each merged group's expected frequency reached at least 5.

5.5 Merged Class Intervals and Chi-Square Calculation

Class Interval (min)	Observed (O)	Expected (E)	$(O - E)^2 / E$
[0.0, 0.5)	17	12.7162	1.4431
[0.5, 1.0)	23	29.3799	1.3854
[1.0, 1.5)	32	29.0894	0.2912
[1.5, 2.0)	21	22.1814	0.0629
[2.0, 2.5)	14	14.8985	0.0542
[2.5, 3.0)	8	9.2829	0.1773
[3.0, 3.5)	7	5.5036	0.4069
≥ 3.5	8	5.8643	0.7778
TOTAL	130	128.9162	4.5988

Table 5.3 — Merged class intervals ($k' = 8$ classes) with Chi-Square contribution per class.

5.6 Test Statistic, Degrees of Freedom, and Decision

$$\text{Chi-Square statistic: } \chi^2_{\text{calc}} = \sum (O - E)^2 / E = 4.5988$$

Degrees of freedom: $df = k' - 1 - m = 8 - 1 - 2 = 5$ ($m = 2$ parameters, shape and scale, estimated from the data)

$$\text{Critical value: } \chi^2_{0.05, 5} = 11.0705$$

$$p\text{-value: } 0.4668$$

Since $\chi^2_{\text{calc}} = 4.5988$ is **less than** the critical value 11.0705, we **fail to reject H_0** at the 5% significance level.

Decision: Fail to Reject H_0 .

5.7 Conclusion

There is not enough statistical evidence, at the 5% significance level, to reject the assumption that ATM service times follow a Gamma distribution with shape $k \approx 2.45$ and scale $\theta \approx 0.65$ minutes (mean service time ≈ 1.60 minutes). The Gamma distribution was also the best-supported choice among the Normal, Gamma, and Lognormal candidates by AIC and K-S criteria, and it is consistent with a service process that has a practical minimum handling time — customers cannot be served in zero time, which the Normal distribution incorrectly allows for. This supports using a Gamma-distributed service process (the "G" in M/G/1, M/G/s, G/G/1, and G/G/s) for this ATM in further queueing simulation work.

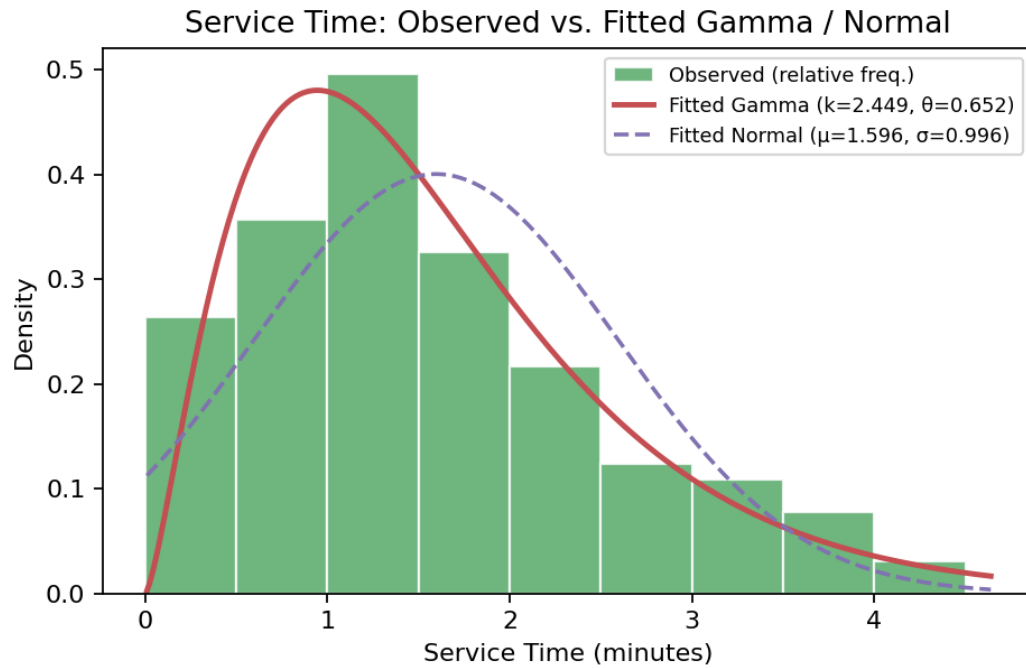


Figure 5.1 — Histogram of observed service times with the fitted Gamma (solid) and Normal (dashed) densities overlaid.

6. Overall Summary and Conclusions

Variable	Fitted Distribution	χ^2 (calc)	df	χ^2 (crit, 0.05)	Decision
Interarrival Time	Exponential($\lambda=0.3023$)	1.5054	2	5.9915	Fail to Reject H0
Service Time	Gamma($k=2.449$, $\theta=0.652$)	4.5988	5	11.0705	Fail to Reject H0

Table 6.1 — Summary of both Chi-Square Goodness-of-Fit tests.

Both tests failed to reject their respective null hypotheses at the 5% significance level. This means the dataset provides no statistical evidence against modeling customer arrivals at this ATM as a Poisson process (Exponential interarrival times) and against modeling transaction durations as Gamma-distributed. Together, these results validate the theoretical distributional assumptions typically used to configure M/M/1, M/M/s, M/G/1, M/G/s, G/G/1, and G/G/s queueing simulations for this ATM: an Exponential arrival process paired with a Gamma-distributed (non-Exponential) service process is the combination best supported by the observed data, rather than a fully Markovian M/M/1-type service assumption.

Note on limitations: the sample size ($n = 130$) is moderate, so the tests have limited power to detect small departures from the fitted distributions, particularly in the tails where classes had to be merged due to low expected frequencies. A larger dataset, or a longer observation window, would allow for finer class intervals and a more powerful test.

7. References

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